TECHTREK • DATA ANALYST FINAL PROJECT

E-Commerce Data Analysis

From Raw Transactions to a Business-Ready Power BI Dashboard

Project Story: Problem → Data → Cleaning → Analysis → Dashboard → Insights → Recommendations

TEAM MEMBERS
Adham Ashraf Mohamed Ahmed
Bassmala Kareem Mohamed
Habiba Elemary
Habiba Shabaan Fares
Mohamed Mohamed Gaber Hassanein

PROJECT LINKS

↗ GitHub Repository

↗ LinkedIn Project Post

The Project Story

One connected workflow, from a business question to a decision-ready dashboard



Problem

Define what the business needs to know

Data

Source a real, relational e-commerce dataset

Cleaning

Prepare 6 tables in Power Query

Analysis

Answer questions with SQL

Dashboard

Build an interactive Power BI report

Insights

Turn numbers into business meaning

Actions

Recommend next steps

This presentation walks through each stage, focusing on what the data revealed, why it matters for the business, and what to do about it.

Business Problem & Objective

Why this analysis?

The business needed to understand how its revenue is actually generated: who buys, what sells, when demand rises or falls, and whether performance depends on a small, fragile slice of customers or products.

Without this, decisions on marketing, inventory and retention were being made without a clear, data-backed picture of performance.

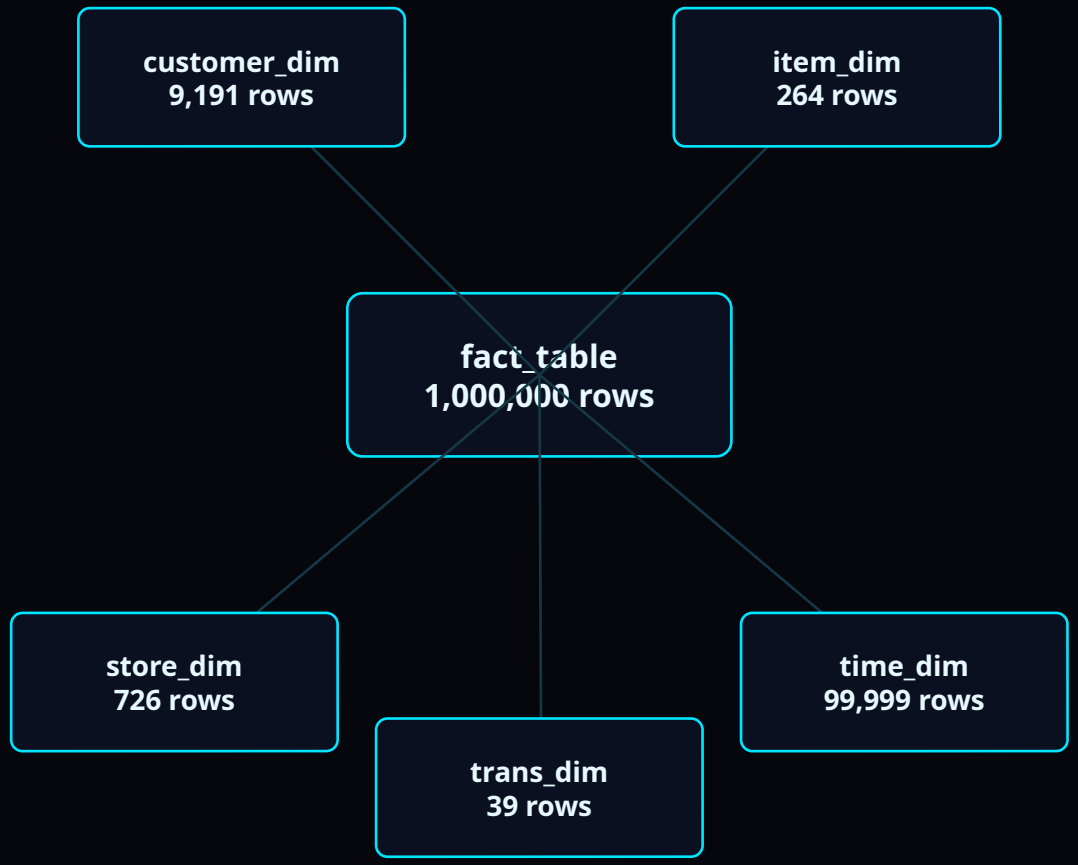
The analysis set out to answer:

- How is revenue distributed across customers — and are we dependent on just a few?
- Is the business growing, and are there seasonal patterns to plan around?
- Which products truly drive revenue, and does volume always mean value?
- What concrete actions can improve retention, product focus, and revenue stability?

The Dataset

A real, relational e-commerce dataset — one fact table and five dimensions

Source: Kaggle — "ECommerce Data Analysis" by mmohaiminulislam



Each dimension links to the fact table through a shared key, enabling revenue and quantity to be sliced by customer, product, store location, date/time, and payment method.

Table	What it holds
fact_table	One row per sale: keys, quantity, unit price, total price
customer_dim	Customer name, contact, national ID
item_dim	Product name, price, country, supplier
store_dim	Division, district, sub-district
time_dim	Hour, day, week, month, quarter, year
trans_dim	Payment type (cash / card / mobile) & bank

Data Cleaning & Preparation

All six tables were cleaned and transformed in Power Query before analysis

1 Missing Values

Filled sensibly instead of deleting rows — e.g. missing units set to the most frequent value "ct"; missing customer names set to "Unknown" to preserve referential integrity.

2 Duplicates

Every table checked for duplicate keys. No duplicate records required removal.

3 Data Types

Keys set to Text, quantities to Whole Number, prices to Decimal Number, dates to Date/Time — preventing formatting and calculation errors.

4 Standardization

Inconsistent unit labels merged (ct. → ct, botllte s → bottles, pk → pack); country names re-capitalized (poland → Poland).

5 Date/Time Rebuild

Raw date-time text (e.g. 01-01-2015 08:14) was parsed with a UK locale and rebuilt into a proper Date/Time field, then validated against hour/day/month/quarter/year.

6 Validation

total_price cross-checked against quantity × unit_price across the full fact table — no inconsistencies found.

Business Questions

The analysis was scoped around three themes

Customer Analysis

- How much does each customer contribute to revenue?
- Are revenues concentrated among a few customers?
- Can customers be segmented meaningfully?
- Which segment drives the most revenue?

Revenue & Growth

- Is the business growing over time?
- How does annual revenue change year to year?
- Are there seasonal patterns to plan around?
- Which months over- or under-perform?

Product Analysis

- Which products generate the most revenue?
- Which products sell the largest quantities?
- Does higher volume always mean higher revenue?
- Is revenue concentrated or well diversified?

SQL Analysis Approach

From descriptive metrics to advanced, window-function-driven analysis

Basic Analysis

- Total revenue, quantity sold, average price
- Total transactions & unique customers
- Revenue by division, district & payment method
- Revenue and transactions by year
- Top customers, top products, peak hours

Core SQL: WHERE · ORDER BY · GROUP BY · JOIN · SUM / AVG / COUNT / MIN / MAX

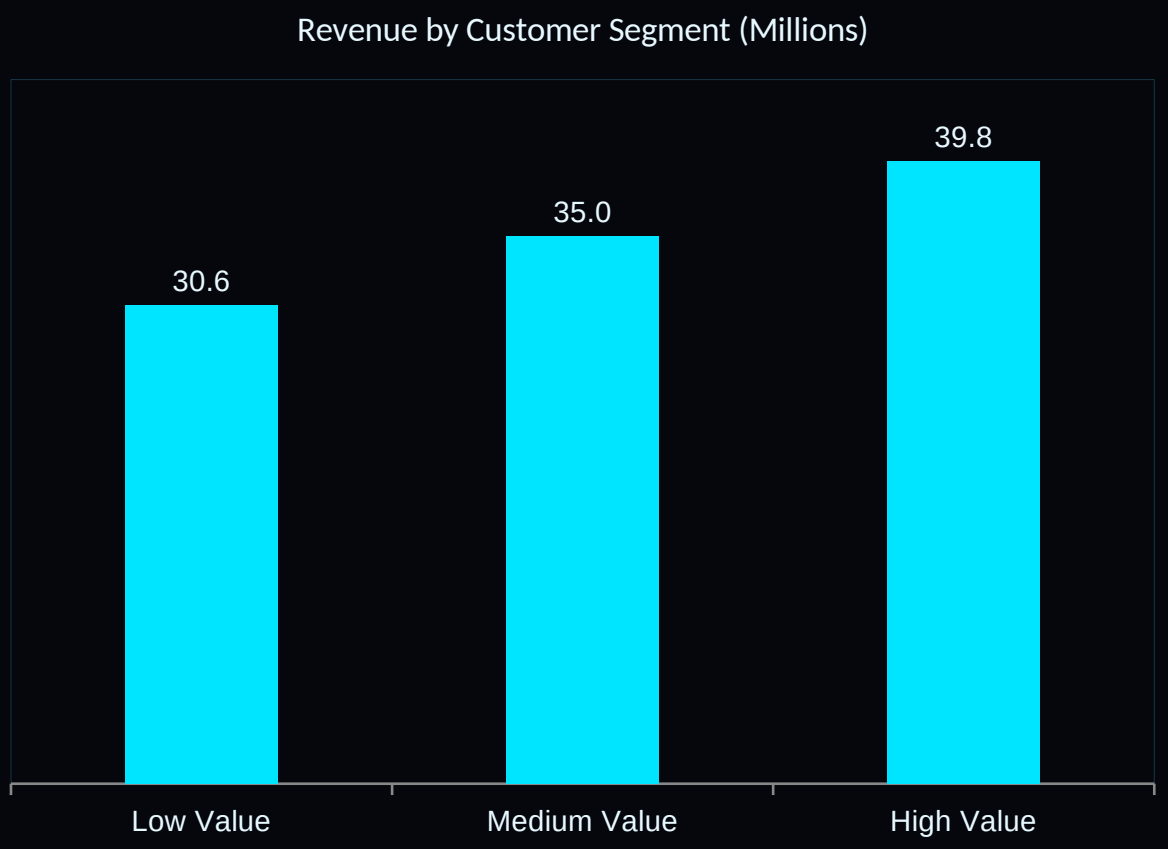
Advanced Analysis

- Customer revenue contribution (window %)
- Customer segmentation with NTILE(3)
- Year-over-year growth with LAG()
- Month-over-month growth with LAG()
- Product performance & revenue concentration

Window functions: OVER() · NTILE() · LAG() · CTEs (WITH)

Result: Customer Segmentation

Customers split into three equal-sized groups by total spend (NTILE)



3,064 / 3,064 / 3,063

Customers per segment (Low / Medium / High)

1.62%

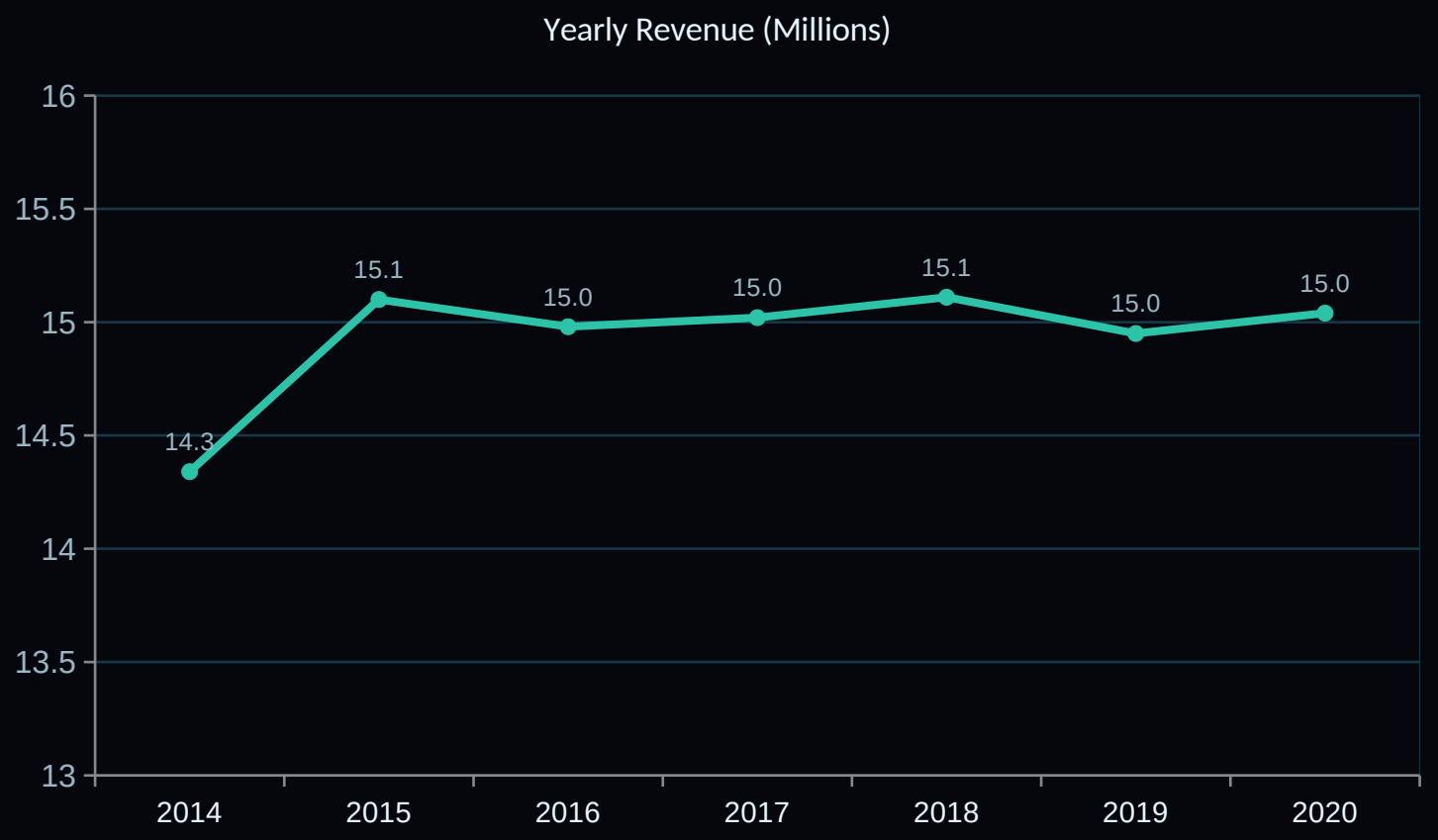
Highest single-customer share of total revenue

Insight

High Value customers are only one-third of the customer base but generate the largest revenue share — and no single customer accounts for more than ~1.6% of revenue, so the business is not dependent on any one buyer.

Result: Revenue Trend & Seasonality

Annual revenue 2014–2020 (2021 excluded — incomplete data)



~15M / yr

Average annual revenue, 2014–2020

Feb ↓

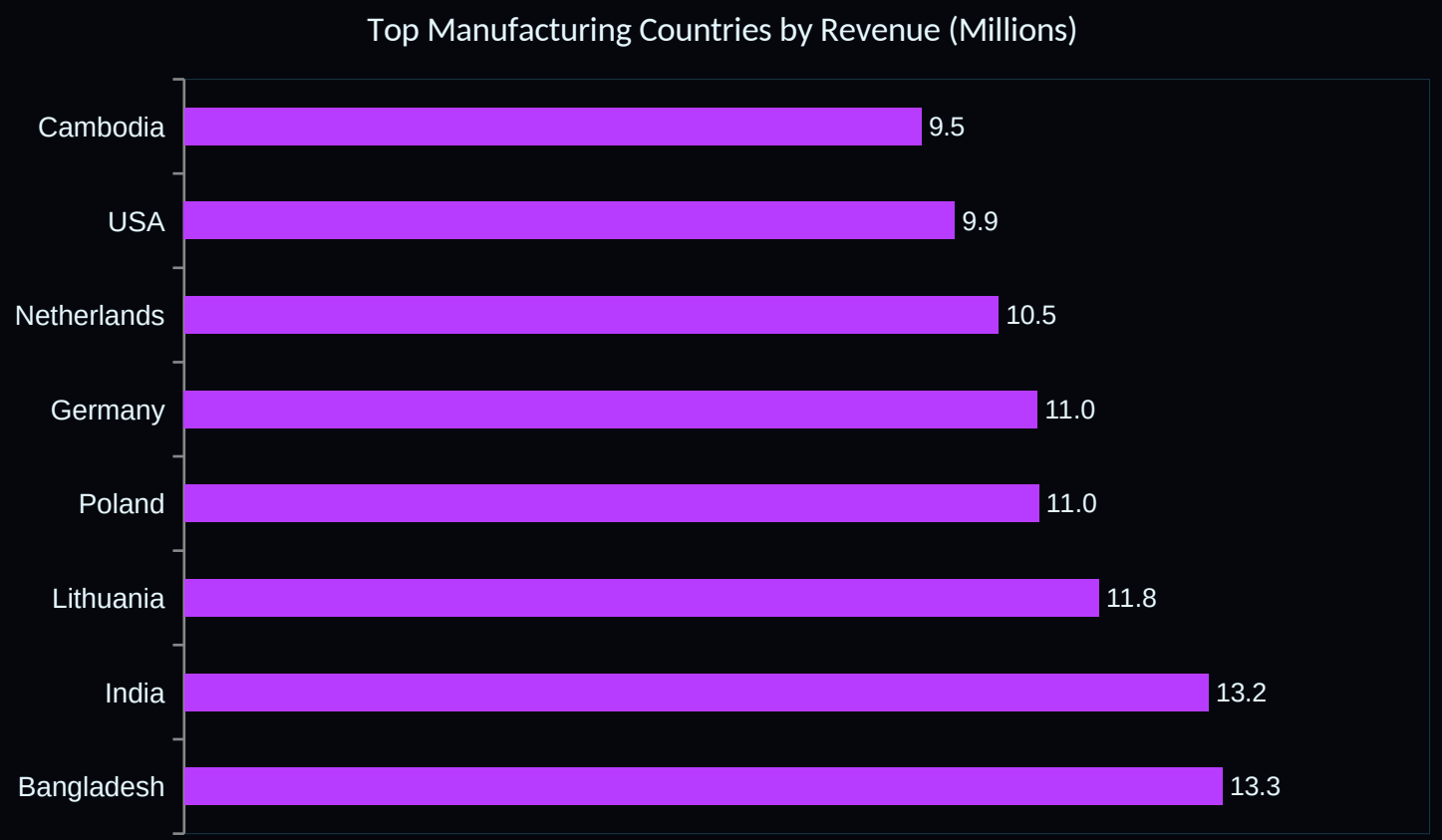
Consistently the weakest month, driven by lower order volume

Insight

Revenue is stable, not rapidly growing — the business is mature. March, May and December consistently outperform, while February needs a seasonal campaign to offset its recurring dip.

Result: Product Performance

Comparing sales volume with revenue generated



1.24%

Top single product's share of total revenue

Coffee & Energy Drinks

Strongest revenue-driving categories

Insight

Higher sales volume does not always mean higher revenue — price matters. Revenue is spread across a broad product portfolio, which reduces the business's dependency on any single product.

The Power BI Dashboard

The live report now uses three connected pages with a consistent cyberpunk / neon visual system.

01

Sales Overview

Core KPIs, YTD trend, division sales and transaction-type mix.

REPORT PAGE

02

Market & Regional Performance

Manufacturing-country funnel, regional map and division performance.

REPORT PAGE

03

Customer & Transaction Insights

Customer KPIs, top-15 customers, customer sales trend and transaction mix.

REPORT PAGE

Sales Overview

1M	105.40M৳	105.40৳	17.79%	2.16%
Total Transactions	Total Sales	Average Transaction Value	YoY %	MoM %

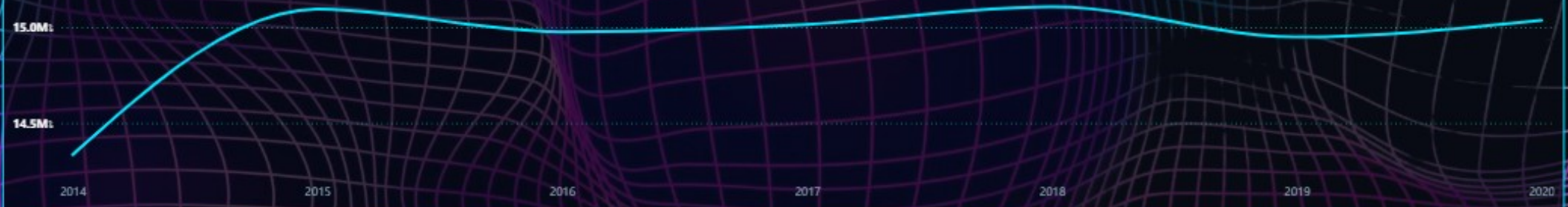
Year

- Select all
- 2014
- 2015
- 2016
- 2017
- 2018
- 2019

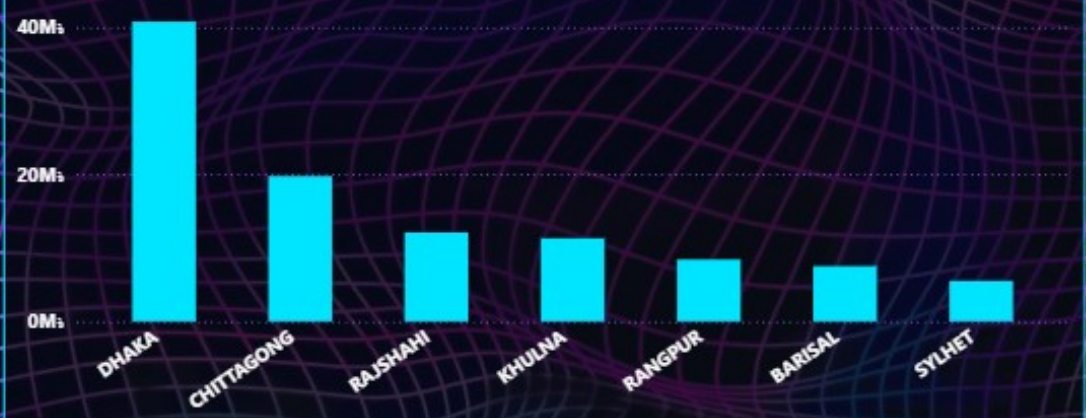
Division

- Select all
- BARISAL
- CHITTAGONG
- DHAKA
- KHULNA
- RAJSHAHI
- RANGPUR

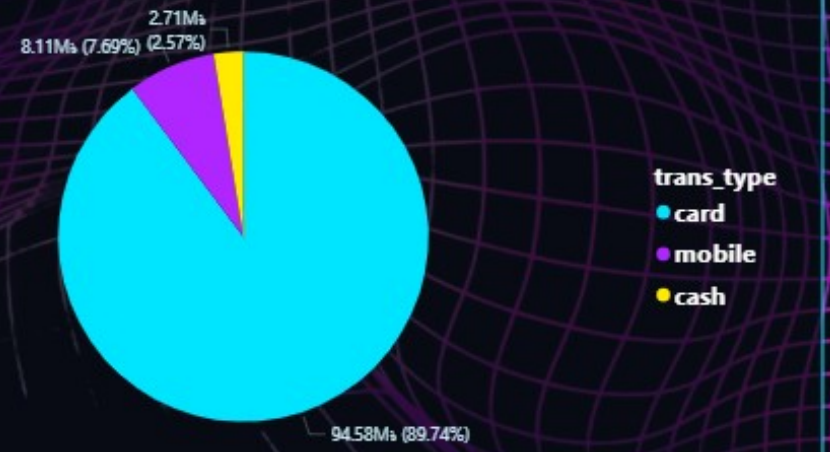
Sales By Year



Total Sales By Division



Total Sales By Transaction Type



Market & Regional Performance

105.40M৳
Total Sales

1M
Total Transactions

Year, Month
- Select all
- 2014
- 2015

Division, District
- Select all
- BARISAL
- CHITTAGONG

Bank, Transaction Type
- Select all
- AB Bank Limited
- Bangladesh Commerce B...

Total Sales By Manufacturing Countries, Suppliers



Total Sales By Division



Total Sales By Division



Customer & Transaction Insights

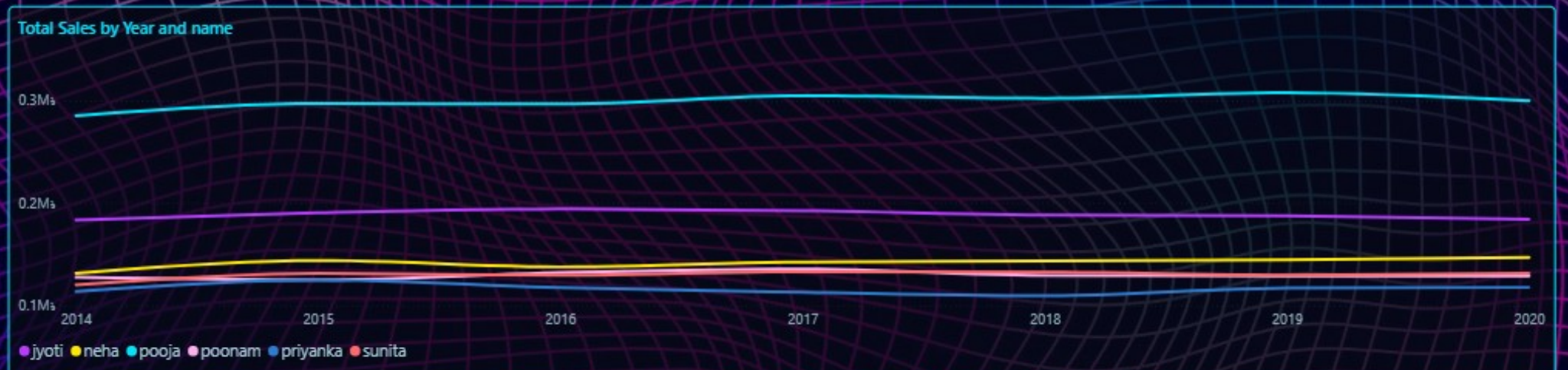
9191	1M	6M	105.40৳	17.79%
Total Customers	Total Transactions	Total Quantity	Average Transaction Value	YoY %

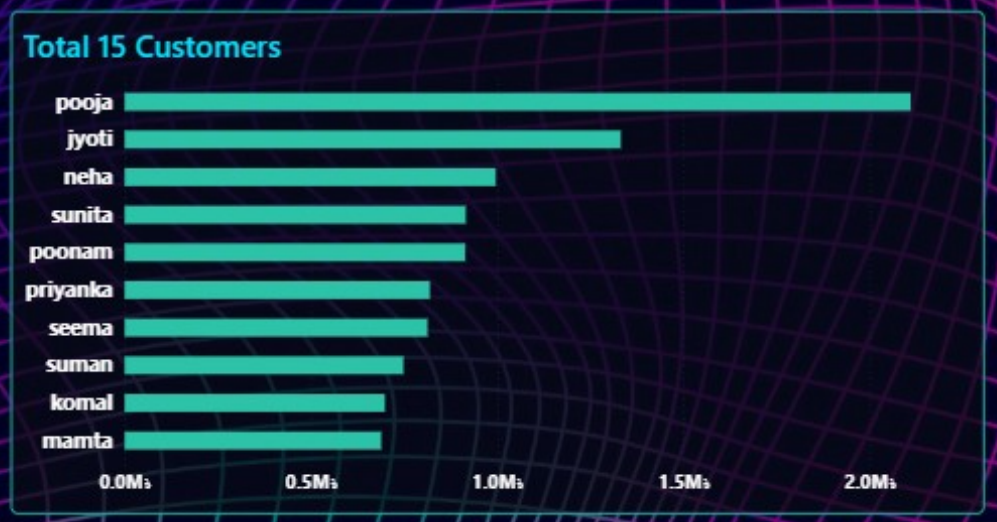
Customer Name (Top 15)
- Select all
- aarti
- jyoti

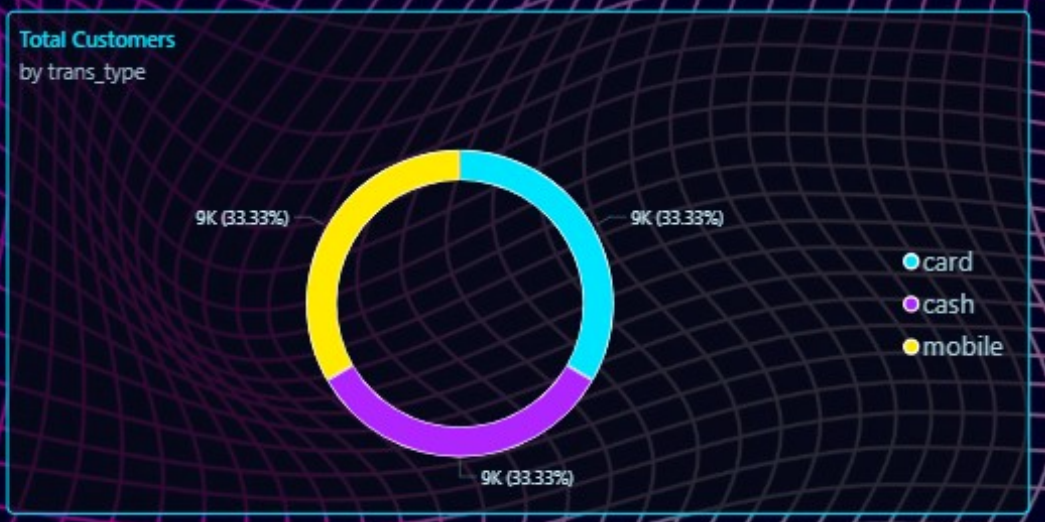
Year, Month
- Select all
- 2014
- 2015

Division, District
- Select all
- BARISAL
- CHITTAGONG

Bank, Transaction Type
- Select all
- AB Bank Limited
- Bangladesh Commerce B...







Key Findings & Insights

Revenue	Revenue held steady at ~15M per year from 2014–2020 — a mature, stable business rather than a rapidly growing one.
Orders	Order volume was similarly stable at ~142,000 orders/year; February is a recurring soft spot driven by fewer orders, not lower prices.
Customers	High Value customers (top third) generate the largest revenue share, yet no single customer exceeds ~1.6% of total revenue.
Products	Coffee and energy-drink products are the strongest revenue drivers; pricing — not just volume — is a major factor in revenue.
Diversification	Revenue is diversified across a broad customer base and product portfolio, limiting dependence on any one customer or product.

Data-Driven Recommendations

Customer

- Loyalty programs for High Value customers
- Personalized promotions for top spenders
- Proactive retention monitoring

Product

- Prioritize inventory for coffee & energy drinks
- Boost marketing on high-revenue products
- Expand cross-selling around top categories

Revenue

- Run seasonal campaigns to offset February dip
- Leverage strong months: March, May, December
- Track monthly revenue for early warning signs

Business

- Keep the diversified product strategy
- Continue broad customer acquisition
- Focus on lifetime value, not just new sign-ups

Limitations & Assumptions

Limitations

- 2021 was excluded from trend analysis — order volume was far lower, indicating incomplete data.
- No profit, cost or margin data — profitability could not be assessed.
- Limited customer demographics restrict deeper profiling.
- No explicit product category field, limiting category-level analysis.
- External factors (promotions, holidays, competitors) are not in the dataset.

Assumptions

- Cleaned records accurately reflect real business transactions.
- Fact-to-dimension relationships are valid throughout.
- Revenue figures are correct — validated against quantity × unit price.
- Cleaning-stage missing-value handling does not materially change conclusions.
- The cleaned data represents normal, typical business operations.

Thank You

TechTrek E-Commerce Analytics — Data Analyst Training Final Project